

Some Mathematics for You

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December 19, 2025

These are some proofs and exercises that I find interesting. I plan to add where I got them from later, but if you want to add anything or have suggestions, and/or questions, feel free to email me at enrojas@cpp.edu.

1 Riemann Integration

Definition 1.1 (lower and upper Riemann sums). Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function and P is a partition $x_0, \dots, x_n$ of $[a, b]$. The lower Riemann sum and upper Riemann sum are defined as

$$L(f, P, [a, b]) = \sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f$$

and

$$U(f, P, [a, b]) = \sum_{j=1}^n (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f,$$

respectively.

Result 1.2. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function and P, P' are partitions of $[a, b]$ such that the list defining P is a sub-list of the list defining P' . Then

$$L(f, P, [a, b]) \leq L(f, P', [a, b]) \leq U(f, P', [a, b]) \leq U(f, P, [a, b]). \quad (1.1)$$

Proof. We first prove the inequality on the left. Suppose that P is the partition $x_0, \dots, x_n$ and P' is the partition $x'_0, \dots, x'_N$ of $[a, b]$, where P is a sub-list of P' . For each $j \in \{1, \dots, n\}$ there exist $k \in \{0, \dots, N-1\}$ and a positive integer m such that

$$x_{j-1} = x'_k < x'_{k+1} < \dots < x'_{k+m} = x_j.$$

We have

$$(x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f = \sum_{i=1}^m (x'_{k+i} - x'_{k+i-1}) \inf_{[x_{j-1}, x_j]} f \leq \sum_{i=1}^m (x'_{k+i} - x'_{k+i-1}) \inf_{[x'_{k+i-1}, x'_{k+i}]} f.$$

Therefore, the sum

$$\sum_{j=1}^n (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f$$

is less than or equal to the sum of sums that correspond to k and m , that is,

$$\sum_{j=1}^N (x'_j - x'_{j-1}) \inf_{[x'_{j-1}, x'_j]} f.$$

Similarly, we have

$$\sum_{i=1}^m (x'_{k+i} - x'_{k+i-1}) \sup_{[x'_{k+i-1}, x'_{k+i}]} f \leq \sum_{i=1}^m (x'_{k+i} - x'_{k+i-1}) \sup_{[x_{j-1}, x_j]} f = (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f.$$

Thus the sum

$$\sum_{j=1}^n (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f$$

is greater than or equal to

$$\sum_{j=1}^N (x'_j - x'_{j-1}) \sup_{[x'_{j-1}, x'_j]} f.$$

This proves the left and right inequalities in (1.1). The middle inequality in (1.1) follows from the fact that the supremum of a set is greater than or equal to the infimum of the set. $\square$

Definition 1.3 (lower and upper Riemann integrals). Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function. The lower Riemann integral and upper Riemann integral of f are defined as

$$L(f, [a, b]) = \sup_P L(f, P, [a, b]) \quad \text{and} \quad U(f, [a, b]) = \inf_P U(f, P, [a, b]),$$

respectively, where the supremum and infimum are taken over all partitions P of $[a, b]$.

Result 1.4. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function. Then

$$L(f, [a, b]) \leq U(f, [a, b]).$$

Definition 1.5 (Riemann integrable). A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is called *Riemann integrable* if

$$L(f, [a, b]) = U(f, [a, b]).$$

In this case, the Riemann integral of f is defined as

$$\int_a^b f = L(f, [a, b]) = U(f, [a, b]).$$

Exercise 1.6. Show that

$$\int_a^b x \, dx = \frac{b^2 - a^2}{2}.$$

Proof. Let P be the partition that divides $[a, b]$ into n subintervals of equal length $\frac{b-a}{n}$. Then the upper sum is

$$U(f, P, [a, b]) = \frac{b-a}{n} \sum_{j=1}^n \left(a + \frac{j(b-a)}{n} \right) = \frac{b^2 - a^2}{2} + \frac{b-a}{2} \cdot \frac{1}{n}.$$

Similarly, the lower sum is

$$L(f, P, [a, b]) = \frac{b-a}{n} \sum_{j=1}^n \left(a + \frac{(j-1)(b-a)}{n} \right) = \frac{b^2 - a^2}{2} - \frac{b-a}{2} \cdot \frac{1}{n}.$$

Thus

$$U(f, [a, b]) \leq \inf_{n \in \mathbb{Z}^+} \left(\frac{b^2 - a^2}{2} + \frac{b-a}{2} \cdot \frac{1}{n} \right) = \frac{b^2 - a^2}{2}$$

and

$$\sup_{n \in \mathbb{Z}^+} \left(\frac{b^2 - a^2}{2} - \frac{b-a}{2} \cdot \frac{1}{n} \right) = \frac{b^2 - a^2}{2} \leq L(f, [a, b]).$$

By the previous result, $L(f, [a, b]) \leq U(f, [a, b])$, so we must have

$$U(f, [a, b]) = L(f, [a, b]) = \frac{b^2 - a^2}{2}.$$

□

Result 1.7 (Cauchy criterion for integrability). *A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ there exists a partition P_n where*

$$U(f, P_n, [a, b]) - L(f, P_n, [a, b]) < \varepsilon.$$

Proof. Suppose that f is Riemann integrable. By definition, for every $\varepsilon > 0$, there exists a partition P_n of $[a, b]$ where

$$L(f, P_n, [a, b]) \geq L(f, [a, b]) - \frac{\varepsilon}{2}.$$

Therefore,

$$U(f, P_n, [a, b]) - L(f, P_n, [a, b]) \leq U(f, [a, b]) - L(f, [a, b]) + \frac{\varepsilon}{2}.$$

Since f is Riemann integrable, then $U(f, [a, b]) = L(f, [a, b])$ and we have that

$$U(f, P_n, [a, b]) - L(f, P_n, [a, b]) \leq \frac{\varepsilon}{2} < \varepsilon.$$

We will prove the claim in the other direction by assuming that for every $\varepsilon > 0$, there exists a partition P_n of $[a, b]$ where

$$U(f, P_n, [a, b]) - L(f, P_n, [a, b]) < \varepsilon.$$

Thus, by definition,

$$U(f, [a, b]) - L(f, [a, b]) \leq U(f, P_n, [a, b]) - L(f, P_n, [a, b]) < \varepsilon.$$

We have that for every $\varepsilon > 0$, $U(f, [a, b]) - L(f, [a, b]) < \varepsilon$, and result 1.4 implies that $U(f, [a, b]) = L(f, [a, b])$.

□

Result 1.8. Suppose that (f_k) is a sequence of Riemann integrable functions $f_k : [a, b] \rightarrow \mathbb{R}$ that converges uniformly to a function $f : [a, b] \rightarrow \mathbb{R}$. Then f is Riemann integrable and

$$\lim_{k \rightarrow \infty} \int_a^b f_k = \int_a^b f.$$

Proof. Let $\varepsilon > 0$. By uniform convergence, there exists $K \in \mathbb{Z}^+$ such that for all $k \geq K$ and all $x \in [a, b]$,

$$|f_k(x) - f(x)| < \frac{\varepsilon}{3(b-a)}.$$

Thus,

$$-\frac{\varepsilon}{3(b-a)} + f_k(x) < f(x) < f_k(x) + \frac{\varepsilon}{3(b-a)},$$

and in particular f is bounded.

By the Cauchy criterion for integrability, there exists a partition P_n such that

$$U(f_k, P_n, [a, b]) - L(f_k, P_n, [a, b]) < \frac{\varepsilon}{3}.$$

From the pointwise inequalities above, we obtain

$$-\frac{\varepsilon}{3(b-a)} + \inf_{[x_{j-1}, x_j]} f_k(x) \leq \inf_{[x_{j-1}, x_j]} f(x) \leq \sup_{[x_{j-1}, x_j]} f(x) \leq \sup_{[x_{j-1}, x_j]} f_k(x) + \frac{\varepsilon}{3(b-a)}.$$

Multiplying each inequality by $(x_j - x_{j-1})$ and summing over j gives

$$-\frac{\varepsilon}{3} + L(f_k, P_n, [a, b]) \leq L(f, P_n, [a, b]) \leq U(f, P_n, [a, b]) \leq U(f_k, P_n, [a, b]) + \frac{\varepsilon}{3}.$$

Therefore,

$$U(f, P_n, [a, b]) - L(f, P_n, [a, b]) \leq U(f_k, P_n, [a, b]) - L(f_k, P_n, [a, b]) + \frac{2\varepsilon}{3} < \frac{\varepsilon}{3} + \frac{2\varepsilon}{3} = \varepsilon.$$

So f satisfies the Cauchy criterion for integrability and is Riemann integrable.

For the convergence of the integrals, let $\varepsilon > 0$. There exists $K \in \mathbb{Z}^+$ such that for all $k \geq K$ and all $x \in [a, b]$,

$$|f_k(x) - f(x)| < \frac{\varepsilon}{b-a}.$$

Then

$$\left| \int_a^b f_k - \int_a^b f \right| = \left| \int_a^b (f_k - f) \right| \leq \int_a^b |f_k - f| < (b - a) \cdot \frac{\varepsilon}{b - a} = \varepsilon.$$

Hence

$$\lim_{k \rightarrow \infty} \int_a^b f_k = \int_a^b f.$$

□

2 Partial differential equations

Result 2.1. Let $\mathbf{F}: U \rightarrow \mathbb{R}^n$ be a C^1 vector field, where $U \subset \mathbb{R}^n$ is open. If

$$\int_V \operatorname{div} \mathbf{F} \, dx = 0$$

for every smooth subdomain $V \Subset U$, then

$$\operatorname{div} \mathbf{F} = 0 \quad \text{in } U.$$

Proof. Let $|B(x, r)|$ denote the Lebesgue measure of $B(x, r) \Subset U$, the open ball centered at x . Let $b \in U$. We have that

$$\frac{1}{|B(b, r)|} \int_{B(b, r)} |\operatorname{div} \mathbf{F} - \operatorname{div} \mathbf{F}(b)| \, dx \leq \sup_{B(b, r)} |\operatorname{div} \mathbf{F} - \operatorname{div} \mathbf{F}(b)|.$$

Since $\operatorname{div} \mathbf{F}$ is continuous, we obtain

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(b, r)|} \int_{B(b, r)} |\operatorname{div} \mathbf{F} - \operatorname{div} \mathbf{F}(b)| \, dx = 0$$

for all $b \in U$. It is worth noting that this is the Lebesgue differentiation theorem where the integrand is continuous. An application of this theorem gives the following:

$$\begin{aligned} \left| \frac{\int_{B(b, r)} \operatorname{div} \mathbf{F} \, dx}{|B(b, r)|} - \operatorname{div} \mathbf{F}(b) \right| &\leq \left| \frac{1}{|B(b, r)|} \int_{B(b, r)} \operatorname{div} \mathbf{F} - \operatorname{div} \mathbf{F}(b) \, dx \right| \\ &\leq \frac{1}{|B(b, r)|} \int_{B(b, r)} |\operatorname{div} \mathbf{F} - \operatorname{div} \mathbf{F}(b)| \, dx. \end{aligned}$$

Hence by the Lebesgue differentiation theorem,

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(b, r)|} \int_{B(b, r)} \operatorname{div} \mathbf{F} \, dx = \operatorname{div} \mathbf{F}(b).$$

Since

$$\int_V \operatorname{div} \mathbf{F} \, dx = 0$$

for every smooth subdomain $V \Subset U$,

$$\operatorname{div} \mathbf{F} = 0$$

for all $b \in U$. □

3 Real Analysis

The following definition can be considered as the kernel of analysis.

Definition 3.1. Let $y \in \mathbb{R}$ and $B \subset \mathbb{R}$. We say that y is *arbitrarily close* to B and write " y acl B " if for every $\varepsilon > 0$, there exists an element $x_\varepsilon \in B$ such that

$$d_{\mathbb{R}}(x_\varepsilon, y) = |x_\varepsilon - y| < \varepsilon.$$

If there exists an $\varepsilon > 0$, such that for every element $x \in B$,

$$d_{\mathbb{R}}(x, z) = |x - z| \geq \varepsilon,$$

then we say z is *away from* B and write " z awf B ".

Result 3.2. If $G = (-\infty, 3140)$ then the number 4200 awf G .

Proof. If we let $\varepsilon = 530$, then for every $x \in G$ we have

$$|x - 4200| = 4200 - x > 4200 - 3140 > 530 = \varepsilon.$$

□

Result 3.3. Consider the set

$$T = [0, 2) \cup \{4 - (1/n) : n \in \mathbb{Z}^+\}.$$

1. Show that 2 acl T .
2. Show that 3 awf T .
3. Show that 4 acl T .

Proof.

1. Let $\varepsilon > 0$. Define $\varepsilon_0 = \min\{1, \varepsilon\}$. Choose

$$x_\varepsilon = 2 - \frac{\varepsilon_0}{2}.$$

Then

$$|x_\varepsilon - 2| = |2 - \frac{\varepsilon_0}{2} - 2| = \frac{\varepsilon_0}{2} < \varepsilon.$$

Therefore, 2 acl T .

2. Let $\varepsilon = \frac{1}{3}$. Note that for $x \in [0, 2)$, we have

$$|x - 3| = 3 - x > 3 - 2 = 1 > \frac{1}{3} = \varepsilon.$$

Let $\varepsilon = 1$. For $x \in \{4 - (1/n) : n \in \mathbb{Z}^+\}$, we have

$$|x - 3| = |4 - \frac{1}{n} - 3| = 1 - \frac{1}{n} > 1 = \varepsilon.$$

Therefore, 3 awf T .

3. Let $\varepsilon > 0$. Choose $n_\varepsilon > \frac{1}{\varepsilon}$ and let

$$x_\varepsilon = 4 - \frac{1}{n_\varepsilon}.$$

Therefore,

$$|x_\varepsilon - 4| = \left| 4 - \frac{1}{n_\varepsilon} - 4 \right| = \frac{1}{n_\varepsilon} < \varepsilon.$$

□

The next proof shows that the supremum of a set is the least upper bound of that set. For that we need the following definition.

Definition 3.4. The *supremum* of a set S is a number $u \in \mathbb{R}$ such that

1. $x \leq u$ for all $x \in S$
2. $u \text{ acl } S$.

Theorem 3.5. Suppose $u \in \mathbb{R}$ is an upper bound for a set $S \subset \mathbb{R}$. Then $u = \sup S$ if and only if u is the least upper bound of S .

Proof. First we assume that u is an upper bound for S and $u = \sup S$. Suppose $y < u$. We need to show that y is not an upper bound for S . Note that $u - y > 0$ and let $\varepsilon_y = u - y$. Since $u \text{ acl } S$, there exists some x_y in S where

$$d_{\mathbb{R}}(x_y, u) = |x_y - u| < \varepsilon_y = u - y.$$

Therefore,

$$u - x_y < u - y \Rightarrow x_y > y$$

which implies that y is not the least upper bound. This shows that u is the least upper bound.

Let $\varepsilon > 0$. Now we assume that u is the least upper bound of S . This implies that $u - \varepsilon$ is not an upper bound for S and there is a real number $x_\varepsilon \in S$ where

$$u - \varepsilon < x_\varepsilon.$$

Therefore,

$$|x_\varepsilon - u| = u - x_\varepsilon < \varepsilon.$$

This implies that $u \text{ acl } S$. This shows that u is the supremum of S . □

Now we prove the "mirror image" of the previous theorem.

Theorem 3.6. Suppose $l \in \mathbb{R}$ is a lower bound for a set $S \subset \mathbb{R}$. Then $l = \inf S$ if and only if l is the greatest lower bound of S .

Proof. First suppose that $l = \inf S$. Suppose $l < y$. Then $y - l > 0$. Let $\varepsilon_y = y - l$. Since l is a lower bound for S , then there exists some x_ε in S such that

$$|x_\varepsilon - l| < \varepsilon_y = y - l$$

which implies that

$$x_\varepsilon - l < y - l \Rightarrow x_\varepsilon < y$$

where the first inequality comes from the definition of the infimum. We have shown that l is the greatest lower bound. Now we assume that l is the greatest lower bound for S . Then $l + \varepsilon$ is not a lower bound and thus there exist an x_ε in S such that

$$l + \varepsilon > x_\varepsilon \Rightarrow |x_\varepsilon - l| < \varepsilon.$$

This implies that l is a lower bound for S and thus the infimum of S . □

4 Functional Analysis

Definition 4.1. Suppose U is a nonempty subset of a normed vector space V and $f \in V$. The *distance* from f to U , denoted $\text{distance}(f, U)$, is defined by

$$\text{distance}(f, U) = \inf\{\|f - g\| : g \in U\}.$$

The next example shows that there is no element of a Banach space that is closest to a closed subspace of that Banach space. Consider the Banach space $C([0, 1])$ with norm $\|g\| = \sup_{[0,1]} |g|$.

Let

$$U = \left\{ g \in C([0, 1]) : \int_0^1 g = 0 \text{ and } g(1) = 0 \right\}.$$

One can verify that this is a subspace of the space $C([0, 1])$. One way to show that this is a closed subspace is to consider the bounded linear functional $\phi : C([0, 1]) \rightarrow \mathbb{R}$ defined by

$$\phi(f) = \int_0^1 f.$$

Since ϕ is a bounded linear functional, then

$$\text{null } \phi = \left\{ f \in C([0, 1]) : \int_0^1 f = 0 \right\}.$$

is a closed subspace of $C([0, 1])$.

Now we consider the bounded linear functional (why is it bounded?) $\psi : C([0, 1]) \rightarrow \mathbb{R}$ defined by

$$\psi(f) = f(1).$$

Since ψ is bounded, then

$$\text{null } \psi = \{f \in C([0, 1]) : f(1) = 0\}$$

is a closed subspace of $C([0, 1])$. Intersecting both null sets implies that the subspace U is closed.

Let $f \in C([0, 1])$ be defined by $f(x) = 1 - x$. For $k \in \mathbb{Z}^+$, let

$$g_k(x) = \frac{1}{2} - x + \frac{x^k}{2} + \frac{x-1}{k+1}.$$

One can verify that $g_k \in U$. Note that for any k , the supremum

$$\sup_{[0,1]} |f(x) - g_k(x)|$$

is obtained as a maximum at $x = 0$. Hence,

$$\lim_{k \rightarrow \infty} \|f - g_k\| = \frac{1}{2}.$$

This implies that $\text{distance}(f, U) \leq \frac{1}{2}$. If $g \in U$, then

$$\int_0^1 (f - g) = \frac{1}{2}$$

and we obtain

$$\frac{1}{2} \leq \sup_{[0,1]} |f - g|$$

by the bounds of the integral. This implies that $\text{distance}(f, U) \geq \frac{1}{2}$. **FINISH PROOF**

5 Measure Theory

Definition 5.1. A set $A \in \mathbb{R}$ is called a *Lebesgue measurable* if there exists a Borel set $B \subset A$ such that the outer measure $|A \setminus B| = 0$.

This means that Lebesgue measurable sets are almost Borel sets (or differ from a Borel set by a set of measure 0). The following is a result that gives equivalence for being a Lebesgue measurable set.

Result 5.2. Suppose $A \subset \mathbb{R}$. Then the following are equivalent:

1. A is a Lebesgue measurable set.
2. For each $\varepsilon > 0$, there exists a closed set $F \subset A$ with $|A \setminus F| < \varepsilon$.
3. There exists closed sets F_k contained in A such that $|A \setminus \bigcup_{k=1}^{\infty} F_k| = 0$.
4. There exists a Borel set $B \subset A$ such that $|A \setminus B| = 0$.
5. For every $\varepsilon > 0$, there exists an open set G where $A \subset G$ and $|G \setminus A| < \varepsilon$.
6. There exists open sets G_k containing A such that $|(\bigcap_{k=1}^{\infty} G_k) \setminus A| = 0$.
7. There exists a Borel set B where $A \subset B$ such that $|B \setminus A| = 0$.

Another result to figure out if A is a Lebesgue measurable set is stated and proven below.

Result 5.3. Suppose $A \subset \mathbb{R}$ and $|A| < \infty$. Then A is a Lebesgue measurable set if and only if for every $\varepsilon > 0$ there exists a set G that is the union of finitely many disjoint bounded open intervals such that $|A \setminus G| + |G \setminus A| < \varepsilon$.

The following theorem states that sequences of measurable functions converge uniformly except on arbitrarily small sets. Before we state and prove the theorem, we should consider an example. Consider the sequence of functions $f_k : [-1, 1] \rightarrow \mathbb{R}$ such that

$$f_k(x) = \begin{cases} 1 & \text{if } x \in [-1, \frac{-1}{k}) \cup (\frac{1}{k}, 1] \\ 2 - k|x| & \text{if } x \in [\frac{-1}{k}, \frac{1}{k}] \end{cases}$$

and consider the function $f : [-1, 1] \rightarrow \mathbb{R}$ defined as

$$f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 2 & \text{if } x = 0. \end{cases}$$

Note that if $x \neq 0$ we can choose $n \geq \frac{1}{|x|}$ so that for all $k \geq n$,

$$|x| \geq \frac{1}{k}$$

and $f_k(x) \rightarrow 1$. If $x = 0$, then $f_k(x) \rightarrow 2$. So we conclude that $f_k(x)$ converges point wise to $f(x)$. To show that we do not have uniform convergence, we use the supremum norm across $[-1, 1]$ and note that

$$\sup_{x \in [-1, 1]} |f_k(x) - f(x)| = \sup_{0 < |x| < \frac{1}{k}} |f_k(x) - f(x)| = \sup_{0 < |x| < \frac{1}{k}} |1 - k|x|| = 1.$$

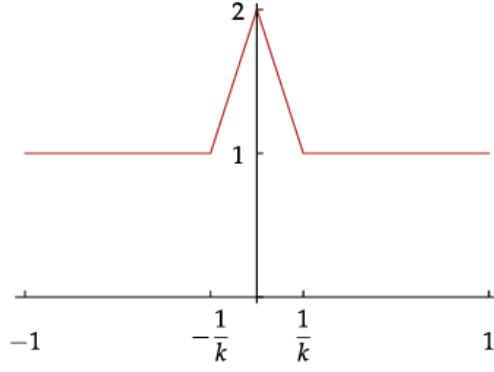


Figure 1: The graph of f_k

Notice that in the set $A = [-1, -\frac{\varepsilon}{2}] \cup (\frac{\varepsilon}{2}, 1]$ we have uniform convergence since that supremum norm taken across A will always obtain 0. Therefore, the "size" of a set where f_k is not uniformly convergent can be made arbitrarily small. With that example, we state the following result.

Theorem 5.4 (Egorov's theorem). *Suppose $(X, \mathcal{S}, \mu)$ is a measure space with $\mu(X) < \infty$ and f_k is a sequence of $\mathcal{S}$ -measurable functions from $X \rightarrow \mathbb{R}$ that converges pointwise on X to a function $f : X \rightarrow \mathbb{R}$. Then for every $\varepsilon > 0$, there exists a set $E \in \mathcal{S}$ such that $\mu(X \setminus E) < \varepsilon$ and f_k converges uniformly to f on E .*

Proof. Let $\varepsilon > 0$ and temporarily fix $n \in \mathbb{Z}^+$. We will first find the set E where f_k converges uniformly to f . Since f is pointwise convergent, then for every $x \in X$ there exists an $N \in \mathbb{Z}^+$ such that

$$|f_k(x) - f(x)| < \frac{1}{n}$$

for all $k \geq N$. Therefore,

$$\bigcup_{m=1}^{\infty} \bigcap_{k=m}^{\infty} \left\{ x \in X : |f_k(x) - f(x)| < \frac{1}{n} \right\} = X.$$

Let $A_{m,n} = \bigcap_{k=m}^{\infty} \left\{ x \in X : |f_k(x) - f(x)| < \frac{1}{n} \right\}$. Then

$$\bigcup_{m=1}^{\infty} A_{m,n} = X.$$

Note that $A_{1,n} \subset A_{2,n} \subset A_{3,n} \subset \dots$. By a well know result from measure theory,

$$\mu(X) = \mu \left(\bigcup_{m=1}^{\infty} A_{m,n} \right) = \lim_{m \rightarrow \infty} \mu(A_{m,n}).$$

By the limit definition, there exists an $m_n \in \mathbb{Z}^+$ such that

$$\mu(X) - \mu(A_{m_n,n}) < \frac{\varepsilon}{2^n}.$$

We now let

$$E = \bigcap_{n=1}^{\infty} A_{m_n,n}.$$

We have that

$$\begin{aligned}
\mu(X \setminus E) &= \mu\left(X \setminus \bigcap_{n=1}^{\infty} A_{m_n, n}\right) \\
&= \mu\left(\bigcup_{n=1}^{\infty} X \setminus A_{m_n, n}\right) \\
&\leq \sum_{n=1}^{\infty} \mu(X \setminus A_{m_n, n}) \\
&= \sum_{n=1}^{\infty} [\mu(x) - \mu(A_{m_n, n})] \\
&< \varepsilon \sum_{n=1}^{\infty} \frac{1}{2^n} = \varepsilon
\end{aligned}$$

which proves the first half of the conclusion in the result. To show uniform convergence in E , we let $\varepsilon' > 0$. Choose $\frac{1}{n} < \varepsilon'$ and note that $E \subset A_{m_n, n}$. Therefore

$$|f_k(x) - f(x)| < \frac{1}{n} < \varepsilon'$$

for all $x \in E$ and for all $k \geq n$. □

Result 5.5. *If $A \subset \mathbb{R}$ and f_k is a sequence of uniformly continuous functions from $A \rightarrow \mathbb{R}$ that converge uniformly to a function $f : A \rightarrow \mathbb{R}$, then f is uniformly continuous on A .*

Proof. Let $\varepsilon > 0$. Since $f_k \rightarrow f$ uniformly, there exists an $n \in \mathbb{Z}^+$ such that

$$|f_n(x) - f(x)| < \frac{\varepsilon}{3}$$

for all $x \in A$. Since f_n is uniformly continuous, then there exists a $\delta > 0$ such that whenever $|x - y| < \delta$ then

$$|f_n(x) - f_n(y)| < \frac{\varepsilon}{3}$$

for all $x, y \in A$. Therefore,

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

proving the result. □

The mathematician Constantine Caratheodory provided us with the following definition and results.

Definition 5.6. Let X be a set and let μ^* be an outer measure on X . A set $E \subset X$ is said to be *measurable with respect to μ^** if for every $A \subset X$, we have

$$\mu^*(E) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$$

where E^c is the complement of E in X .

Theorem 5.7. *Let X be a set and let μ^* be an outer measure on X . The collection of sets that are measurable with respect to μ^* is a σ -algebra.*

The next result tells us that the average difference between a function and a specific function value at b is small on small intervals centered at b . This is known as the classical result, the Lebesgue differentiation theorem.

Theorem 5.8. *Let $f \in L^1(\mathbb{R})$, then*

$$\lim_{t \rightarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| = 0$$

for almost every $b \in \mathbb{R}$.

Proof. The result holds if f is continuous at b due to the estimate (why?)

$$\frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| \leq \sup_{x \in [b-t, b+t]} |f(x) - f(b)|.$$

For an arbitrary function $f \in L^1(\mathbb{R})$ we let $\delta > 0$. From a result in measure theory associated with the L^1 norm, for $f \in L^1(\mathbb{R})$ and for each $k \in \mathbb{Z}^+$ there exists a continuous function $g_k : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\|f - g_k\|_1 < \frac{\delta}{k2^k}. \quad (5.1)$$

Let

$$B_k = \left\{ b \in \mathbb{R} : |f(b) - g_k(b)| \leq \frac{1}{k} \text{ and } M(f - g_k)(b) \leq \frac{1}{k} \right\} \quad (5.2)$$

where M is the Hardy-Littlewood maximal function.

Then,

$$\mathbb{R} \setminus B_k = \left\{ b \in \mathbb{R} : |f(b) - g_k(b)| > \frac{1}{k} \right\} \cup \left\{ b \in \mathbb{R} : M(f - g_k)(b) > \frac{1}{k} \right\}.$$

From (5.1) and using Chebyshev's inequality and the Hardy-Littlewood maximal inequality, we obtain that the outer measure

$$\left| \left\{ b \in \mathbb{R} : |f(b) - g_k(b)| > \frac{1}{k} \right\} \right| < \frac{\delta}{2^k}$$

and

$$\left| \left\{ b \in \mathbb{R} : M(f - g_k)(b) > \frac{1}{k} \right\} \right| < \frac{3\delta}{2^k}$$

respectively. This implies that

$$|\mathbb{R} \setminus B_k| < \frac{\delta}{2^{k-1}}.$$

Let

$$B = \bigcap_{k=1}^{\infty} B_k.$$

Since

$$\mathbb{R} \setminus B = \mathbb{R} \setminus \bigcap_{k=1}^{\infty} B_k = \bigcup_{k=1}^{\infty} (\mathbb{R} \setminus B_k),$$

then

$$|\mathbb{R} \setminus B| = \left| \bigcup_{k=1}^{\infty} (\mathbb{R} \setminus B_k) \right| \leq \sum_{k=1}^{\infty} |\mathbb{R} \setminus B_k| < \sum_{k=1}^{\infty} \frac{\delta}{2^{k-1}}$$

The last sum on the inequality converges to $\frac{1}{2}$, therefore

$$|\mathbb{R} \setminus B| < 4\delta. \quad (5.3)$$

Now, let $b \in B$ and $t > 0$. Then for each $k \in \mathbb{Z}^+$,

$$\begin{aligned} \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| &\leq \frac{1}{2t} \left(\int_{b-t}^{b+t} |f - g_k| + \int_{b-t}^{b+t} |g_k - g_k(b)| + \int_{b-t}^{b+t} |g_k(b) - f(b)| \right) \\ &\leq M(f - g_k) + \frac{1}{2t} \int_{b-t}^{b+t} |g_k - g_k(b)| + \frac{1}{k} \\ &\leq \frac{2}{k} + \frac{1}{2t} \int_{b-t}^{b+t} |g_k - g_k(b)| \end{aligned}$$

where the second inequality comes from (5.2). Since g_k is continuous at b , for each $k \in \mathbb{Z}^+$ there exists a $\delta_k > 0$ such that

$$|g_k(x) - g_k(b)| < \frac{1}{k} \quad \text{whenever} \quad |x - b| < \delta_k.$$

Hence, for each $k \in \mathbb{Z}^+$,

$$\frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| \leq \frac{2}{k} + \frac{1}{2t} \int_{b-t}^{b+t} |g_k - g_k(b)| < \frac{3}{k}$$

for all $0 < t < \delta_k$.

We conclude that for all $b \in B$

$$\lim_{t \rightarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| = 0. \quad (5.4)$$

Now let A be the set of numbers $a \in \mathbb{R}$ such that

$$\lim_{t \rightarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| > 0$$

or does not exist. By (5.3) and (5.4), we have that $A \subset \mathbb{R} \setminus B$ and

$$|A| \leq |\mathbb{R} \setminus B| < 4\delta$$

Since δ is an arbitrary positive number, we have that the outer measure $|A| = 0$ and thus the claim (5.4) holds for almost every $b \in \mathbb{R}$. $\square$

WRITE THE SECOND VERSION OF LDT

6 Fractal Geometry

Result 6.1. Let $(\mathbb{R}^m, d_{m,p})$ be the metric space where, for $x, y \in \mathbb{R}^m$ and $p \geq 1$,

$$d_{m,p}(x, y) = \left(\sum_{j=1}^m |x_j - y_j|^p \right)^{1/p}.$$

Then $(\mathbb{R}^m, d_{m,p})$ is complete.

Proof. Let $x^{(n)} = (x_1^{(n)}, \dots, x_m^{(n)})$ be a Cauchy sequence in $\mathbb{R}^m$. By definition, for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{Z}^+$ such that for all $l, k \geq n_0$,

$$\left(\sum_{j=1}^m |x_j^{(l)} - x_j^{(k)}|^p \right)^{1/p} < \varepsilon.$$

In particular, for each fixed $i \in \{1, \dots, m\}$,

$$|x_i^{(l)} - x_i^{(k)}|^p \leq \sum_{j=1}^m |x_j^{(l)} - x_j^{(k)}|^p < \varepsilon^p.$$

Hence the sequence $(x_i^{(n)})$ is Cauchy in $\mathbb{R}$ for each i , and since $\mathbb{R}$ is complete, there exists $x_i \in \mathbb{R}$ such that $x_i^{(n)} \rightarrow x_i$ as $n \rightarrow \infty$.

Define $x = (x_1, \dots, x_m) \in \mathbb{R}^m$. Let $\varepsilon > 0$. For each i there exists $N_i \in \mathbb{Z}^+$ such that for all $n \geq N_i$,

$$|x_i^{(n)} - x_i| < \frac{\varepsilon}{m^{1/p}}.$$

Let $N = \max\{N_1, \dots, N_m\}$. Then for all $n \geq N$,

$$\sum_{j=1}^m |x_j^{(n)} - x_j|^p < m \left(\frac{\varepsilon}{m^{1/p}} \right)^p = \varepsilon^p.$$

Therefore,

$$d_{m,p}(x^{(n)}, x) = \left(\sum_{j=1}^m |x_j^{(n)} - x_j|^p \right)^{1/p} < \varepsilon.$$

So $x^{(n)} \rightarrow x$ in $(\mathbb{R}^m, d_{m,p})$, and the space is complete. □

Theorem 6.2 (Finite Intersection Property). *Let $X = \mathbb{R}^m$ and (X, d) be a metric space. Let $\{K_\lambda\}_{\lambda \in \Lambda}$ be a collection of compact subsets $K_\lambda \subset X$ such that the finite intersection of every subcollection of $\{K_\lambda\}_{\lambda \in \Lambda}$ is non-empty. Then*

$$\bigcap_{\lambda \in \Lambda} K_\lambda \neq \emptyset.$$

Proof. We prove the theorem by contraposition. Suppose

$$\bigcap_{\lambda \in \Lambda} K_\lambda = \emptyset.$$

Fix some $\lambda_0 \in \Lambda$. Since each K_λ is compact, and $X = \mathbb{R}^m$, by the Heine–Borel theorem each K_λ is closed and bounded. In particular, each complement K_λ^c is open, and the collection $\{K_\lambda^c\}_{\lambda \in \Lambda}$ is an open cover of K_{λ_0} .

Since K_{λ_0} is compact, there exists a finite subcover:

$$K_{\lambda_0} \subset \bigcup_{j=1}^q K_{\lambda_j}^c$$

for some $q \in \mathbb{Z}^+$ and $\lambda_1, \dots, \lambda_q \in \Lambda$. This means

$$\bigcap_{j=0}^q K_{\lambda_j} = \emptyset,$$

where by convention we include λ_0 among the indices. (the set K_{λ_0} is intersected with sets that do not contain it). Thus we have found a finite subcollection whose intersection is empty, which is the contrapositive of the statement in the theorem. Therefore the original claim holds. $\square$

Definition 6.3. (Lipschitz Constant and Lipschitz Continuous) Let $f : X \rightarrow Y$, where (X, d_X) and (Y, d_Y) are metric spaces. The *Lipschitz constant* of f , denoted by $\text{Lip}(f)$ is given as follows,

$$\text{Lip}(f) = \inf\{r \geq 0 : d_Y(f(x), f(y)) \leq r \cdot d_X(x, y) \text{ where } x, y \in X\}.$$

Moreover, a function is said to be *Lipschitz continuous* if and only if there exists an $M > 0$ such that

$$d_Y(f(x), f(y)) \leq M \cdot d_X(x, y)$$

for all $x, y \in X$.

Result 6.4. A function that is Lipschitz continuous is uniformly continuous.

Proof. Suppose that f is Lipschitz continuous. Therefore,

$$d_Y(f(x), f(y)) \leq M \cdot d_X(x, y)$$

where $M > 0$. Let $\varepsilon > 0$ and $x, y \in X$ where $d_X(x, y) < \frac{\varepsilon}{M}$. Then,

$$d_Y(f(x), f(y)) \leq M \cdot d_X(x, y) < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

This implies that f is uniformly continuous. $\square$

Result 6.5. *A Lipschitz function sends a Cauchy sequence of (X, d_X) onto a Cauchy sequence of (Y, d_Y) .*

Proof. Let f be a Lipschitz continuous function and $\{x_n\}_{n=1}^\infty$ be a Cauchy sequence. Let $\varepsilon > 0$. Then there exists an $N \in \mathbb{Z}^+$ such that for $m, n \geq N$,

$$d_X(x_m, x_n) < \frac{\varepsilon}{M}.$$

where M is the constant from the definition a Lipschitz continuous function. Since f is Lipschitz continuous,

$$d_Y(f(x_m), f(x_n)) \leq M \cdot d_X(x_m, x_n) < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

This implies that the sequence $\{f(x_n)\}_{n=1}^\infty$ is a Cauchy sequence. $\square$

Result 6.6 (The contraction mapping principle). *Let (X, d) be a complete metric space. If $\psi : X \rightarrow X$ is a contraction on X with scaling ration $r_\psi = r$, then there is a unique $x^* \in X$, called a fixed point of ψ such that*

$$\psi(x^*) = x^*.$$

Moreover, given any initial point $x_0 \in X$ the sequence $\{x_n\}_{n=0}^\infty$ given by

$$x_n := \psi(x_{n-1})$$

converges to x^ .*

Before we prove this theorem, we will exemplify a contraction. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ where $a > 1$,

$$f(x) = \frac{1}{a} \sin(x).$$

By the mean value theorem, for any $x, y \in \mathbb{R}$, we have

$$|f(x) - f(y)| = |f'(c)| |x - y| \leq \frac{1}{a} |x - y|$$

where $c \in \mathbb{R}$. Therefore, $f(x)$ is a contraction on $\mathbb{R}$ with a scaling ratio $\frac{1}{a}$.

It is interesting how the first part of this proof requires us to show that the sequence $\{x_n\}_{n=0}^\infty$ is Cauchy and therefore converges to a point in X . Then we use the continuity of the map, through Lipschitz continuity, to show that the point is fixed through the map.

Proof. Let $x_0 \in X$. Let $m, n \in \mathbb{N}$ and, without loss of generality, assume $m < n$. By the triangle inequality

$$d(x_m, x_n) \leq d(x_m, x_{m+1}) + d(x_{m+1}, x_{m+2}) + \dots + d(x_{n-1}, x_n).$$

Since ψ is a contraction with scaling ratio $0 < r < 1$, we have

$$d(x_k, x_{k+1}) = d(\psi(x_{k-1}), \psi(x_k)) \leq r \cdot d(x_{k-1}, x_k)$$

for any $k \in \mathbb{Z}^+$. This is because we can write $\psi(x_{k-1}) = x_k$ and $\psi(x_k) = x_{k+1}$.

We use induction. Let $k = 1$ to obtain

$$d(x_1, x_2) = d(\psi(x_0), \psi(x_1)) \leq r \cdot d(x_0, x_1) = r \cdot d(x_0, \psi(x_0)).$$

The inductive step requires us to assume that for $k > 1$, we have

$$d(x_k, x_{k+1}) \leq r^k \cdot d(x_0, \psi(x_0)).$$

Note that

$$d(x_{k+1}, x_{k+2}) < r^{k+1} \cdot d(x_0, \psi(x_0)) < r^k \cdot d(x_0, \psi(x_0)).$$

This induction shows that

$$d(x_k, x_{k+1}) \leq r^k \cdot d(x_0, \psi(x_0))$$

for all $k \in \mathbb{Z}^+$.

Therefore, using the triangle inequality one more time, for all $m, n \in \mathbb{N}$ we obtain

$$d(x_m, x_n) \leq (r^m + r^{m+1} + \dots + r^{n-1}) \cdot d(x_0, \psi(x_0)).$$

Since the scaling ratio $0 < r < 1$, the series

$$r^m + r^{m+1} + \dots + r^{n-1} = r^m \cdot \left(\frac{1 - r^{n-m}}{1 - r} \right) \leq \frac{r^m}{1 - r}.$$

Hence for all integers $m \geq 0$ and $n \geq 1$, we have

$$d(x_m, x_n) \leq \frac{r^m}{1 - r} \cdot d(x_0, \psi(x_0)).$$

This implies that $\{x_n\}_{n=0}^\infty$ is a Cauchy sequence in X . Since X is complete, it follows that the sequence $\{x_n\}_{n=0}^\infty$ has a limit. Let us denote that limit by $x^* \in X$. It follows that

$$x^* = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \psi(x_{n-1}).$$

Since ψ is a contraction, it is Lipschitz continuous, and we have

$$x^* = \lim_{n \rightarrow \infty} \psi(x_{n-1}) = \psi \left(\lim_{n \rightarrow \infty} x_{n-1} \right) = \psi(x^*).$$

This proves the convergence of the sequence. We will not show the uniqueness, however, it is found through the definition of the metric d .

□

We can use the contraction mapping principle to prove the following *Picard's Theorem*.

Theorem 6.7. *Let $x_0 \in \mathbb{R}$ and $\mathbf{y}_0 \in \mathbb{R}^m$, and let $[a, b]$ and B be compact neighborhoods of x_0 and $\mathbf{y}_0$ in $\mathbb{R}$ and $\mathbb{R}^m$, respectively, such that the function $\phi : [a, b] \times B \rightarrow \mathbb{R}^m$ is jointly continuous and Lipschitz with respect to the second variable, in the following sense: There exists $M > 0$ such that*

$$|\phi(x, \mathbf{y}) - \phi(x, \mathbf{z})| \leq M \cdot |\mathbf{y} - \mathbf{z}|.$$

Then there exists a nonempty open interval I centered at x_0 such that the initial-value problem $\mathbf{y}' = \phi(x, \mathbf{y})$, $\mathbf{y}(x_0) = \mathbf{y}_0$, has a unique solution on I , denoted $\mathbf{y} = \mathbf{y}(x)$.

Proof. We show that for a small enough $\delta > 0$. and if we take the interval J_δ to be of the form $J_\delta := [x_0 - \delta, x_0 + \delta] \subseteq (a, b)$, the (typically nonlinear) integral operator $L : X \rightarrow X$, given by

$$L(\mathbf{y})(x) := \mathbf{y}_0 + \int_{x_0}^x \phi(t, \mathbf{y}(t)) dt$$

for all $x \in J_\delta$, is well defined and a contraction. Also, X is the space $C(J_\delta)$ equipped with the supremum norm $\|\cdot\|_\infty$.

The operator is well defined because it includes an integral which returns a continuous function for the continuous integrand $\phi(t, y(t))$, hence $L(\mathbf{y}) \in C(J_\delta)$ for all $x \in J_\delta$.

Choose $\delta < b = \min \{|x_0 - a|, |x_0 - b|, \frac{1}{M}\}$. We have that

$$\begin{aligned} |L(\mathbf{y})(x) - L(\mathbf{z})(x)| &= \left| \int_{x_0}^x \phi(t, \mathbf{y}(t)) dt - \int_{x_0}^x \phi(t, \mathbf{z}(t)) dt \right| \\ &= \left| \int_{x_0}^x \phi(t, \mathbf{y}(t)) - \phi(t, \mathbf{z}(t)) dt \right| \\ &\leq |x - x_0| \cdot \sup_{t \in J_\delta} |\phi(t, \mathbf{y}(t)) - \phi(t, \mathbf{z}(t))| \\ &\leq |x - x_0| \cdot M \cdot \sup_{t \in J_\delta} |\mathbf{y}(t) - \mathbf{z}(t)| \\ &\leq \delta \cdot M \cdot \sup_{t \in J_\delta} |\mathbf{y}(t) - \mathbf{z}(t)| \\ &< b \cdot M \cdot \|\mathbf{y} - \mathbf{z}\|_\infty \end{aligned}$$

where the fourth inequality comes from the Lipschitz condition. Therefore, taking the supremum, we obtain

$$\|L(\mathbf{y}) - L(\mathbf{z})\|_\infty = \sup_{x \in J_\delta} |L(\mathbf{y})(x) - L(\mathbf{z})(x)| < b \cdot M \cdot \|\mathbf{y} - \mathbf{z}\|_\infty.$$

Since $b \cdot M$ is less than 1, we have that L is a contraction with scaling ration $b \cdot M$. By the contraction mapping principle, there exists a unique $\mathbf{y}^* \in C(J_\delta)$ such that

$$L(\mathbf{y}^*)(x) = \mathbf{y}^*(x)$$

for all $x \in J_\delta$. Note that

$$\mathbf{y}^*(x) = \mathbf{y}_0 + \int_{x_0}^x \phi(t, \mathbf{y}^*(t)) dt$$

satisfies the IVP. □

Result 6.8. *Let $A \in \mathbb{K}(\mathbb{R}^m)$ and $\varepsilon > 0$. Then there exists a self-similar system $\Phi = \{\phi_j\}_{j=1}^N$ with attractor F_Φ such that*

$$d_H(A, F_\Phi) < \varepsilon.$$

Proof. Let $\varepsilon > 0$. Since A is compact then it can be expressed as subset of a finite open cover composed of open balls B_j with radii at most $\frac{\varepsilon}{4}$, more specifically,

$$A \subset \bigcup_{j=1}^N B_j.$$

FINISH PROOF □

7 Algebraic Topology

A *path* in a space X is a continuous map $f : I \rightarrow X$ where I is the unit interval $[0, 1]$.

Definition 7.1. A *homotopy* of paths in X is a family $f_t : I \rightarrow X$, $0 \leq t \leq 1$, such that

1. The endpoints $f_t(0) = x_0$ and $f_t(1) = x_1$ are independent of t .
2. The associated map $F : I \times I \rightarrow X$ defined by $F(s, t) = f_t(s)$ is continuous.

For example, any two paths f_0 and f_1 in $\mathbb{R}^n$ (or a convex subset of $\mathbb{R}^n$) having the same endpoints x_0 and x_1 are *homotopic* via the homotopy

$$f_t(s) = (1 - t)f_0(s) + tf_1(s).$$

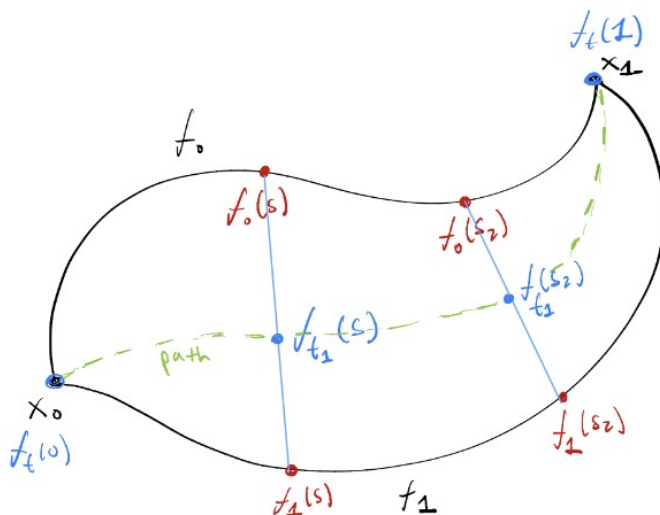


Figure 2: Linear homotopy in $\mathbb{R}^2$

Result 7.2. The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.

Proof.

1. *Reflexivity:* The constant homotopy $f_t = f$ shows that $f \simeq f$.
2. *Symmetry:* If $f_0 \simeq f_1$ via f_t , then $f_1 \simeq f_0$ via the inverse homotopy f_{1-t} .
3. *Transitivity:* If $f_0 \simeq f_1$ via f_t and if $f_1 = g_0$ and $g_0 \simeq g_1$ via g_t , then $f_0 \simeq g_1$ via the homotopy h_t

$$h_t(x) = \begin{cases} f_{2t}(x) & \text{if } 0 \leq t \leq \frac{1}{2} \\ g_{2t-1}(x) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

□

The equivalence class of a path f under the equivalence relation of homotopy is denoted $[f]$ and referred to as the *homotopy class of f* .

Given two paths $f, g : I \rightarrow X$ such that $f(1) = g(0)$, there is a *composition product path* $f \cdot g$ that first traverses f and then g , defined by

$$f \cdot g(s) \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ g(2s - 1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$$

We restrict our attention to paths $f : I \rightarrow X$ with the same starting and ending point $f(0) = f(1) = x_0 \in X$. Such paths are called *loops* and x_0 is the *basepoint*.

The set of all homotopy classes $[f]$ of loops at the base point x_0 is denoted $\pi_1(X, x_0)$.

Result 7.3. $\pi_1(X, x_0)$ is a group with respect to the product $[f][g] = [f \cdot g]$. This group is known as the **fundamental group**.

As an example, for a convex set X in $\mathbb{R}^n$ with base point $x_0 \in X$, we have $\pi_1(X, x_0) = 0$. This is shown by the homotopy

$$H(t, s) = (1 - t)f_0(s) + tx_0.$$

In the figure below, X is the torus, and the red path belongs to one class of homotopies while the blue path belongs to another class.

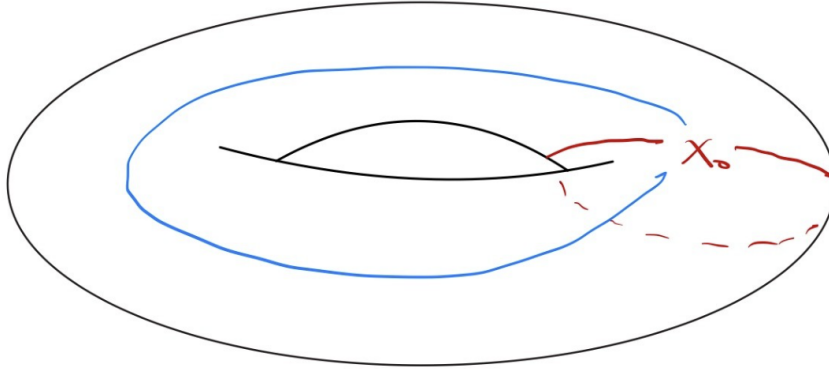


Figure 3: The space which is the torus

It turns out that if your space X is path-connected, the group $\pi_1(X, x_0)$ is independent of the base point (up to an isomorphism). Thus, we can write the group as $\pi_1(X)$.

Result 7.4. A space X is simply-connected if and only if $\pi_1(X) = 0$.

In the next proof we make use of the fact that homotopies are preserved under images of continuous maps. The result shows that the fundamental group of the circle is isomorphic to $\mathbb{Z}$.

Result 7.5. *The map $\Phi : \mathbb{Z} \rightarrow \pi_1(S^1)$ sending an integer n to the homotopy class of the loop $\omega_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$ based at $(1, 0)$ is an isomorphism.*

Proof. We use the map $p : \mathbb{R} \rightarrow S^1$ given by $p(s) = (\cos 2\pi s, \sin 2\pi s)$. You can think of this map as wrapping $\mathbb{R}$ around the circle and integers correspond to full loops.

We also consider the helix $r(s) = (\cos 2\pi s, \sin 2\pi s, s)$, whose projection is the map p and paths on this helix are thought of as paths in $\mathbb{R}$.

Next we define a map $\tilde{\omega}_n : [0, 1] \rightarrow \mathbb{R}$ where $\tilde{\omega}_n(s) = ns$ which is a path from 0 to n .

So, the composition $r(\tilde{\omega}_n(s)) = (\cos 2\pi ns, \sin 2\pi ns, ns)$ wraps around the helix up to a height n and around it $|n|$ times. Meanwhile, the composition $p(\tilde{\omega}_n(s)) = (\cos 2\pi ns, \sin 2\pi ns) = \omega_n(s)$ wraps around the circle $|n|$ times. (Remember that $s \in [0, 1]$ for these compositions)

We call $\tilde{\omega}_n(s)$ the *lift* of ω_n .

Note if $\tilde{f}$ is a path from 0 to n , then it is homotopic to $\tilde{\omega}_n$ via the linear interpolation homotopy $(1-t)\tilde{f} + t\tilde{\omega}_n$, thus its image $p(\tilde{f})$ is homotopic to the image $p(\tilde{\omega}_n) = \omega_n$. Thus paths on the helix (or circle) are homotopic to paths determined by ω_n . With this, we can define the map $\Phi(n)$ as the map that sends n to any class of paths from 0 to n , not just ω_n .

Now we show that Φ is a homomorphism. Let $\tau_m : \mathbb{R} \rightarrow \mathbb{R}$ be the translation where $\tau_m(x) = x + m$. We have

$$\tilde{\omega}_m \cdot (\tau_m(\tilde{\omega}_n)) = \begin{cases} 2ms & \text{if } 0 \leq s \leq \frac{1}{2} \\ n(2s-1) + m & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$$

which is a path in $\mathbb{R}$ from 0 to $n + m$. Therefore, $\Phi(n + m)$ is the homotopy class of the loop in S^1 that is the image

$$p(\tilde{\omega}_m \cdot (\tau_m(\tilde{\omega}_n))) = \begin{cases} (\cos 4\pi ms, \sin 4\pi ms) & \text{if } 0 \leq s \leq \frac{1}{2} \\ (\cos 2\pi[(2s-1)n + m], \sin 2\pi[(2s-1)n + m]) & \text{if } \frac{1}{2} \leq s \leq 1. \end{cases}$$

That image is equal to $\omega_m \cdot \omega_n$. Therefore $\Phi(n + m)$ sends $m + n$ to the homotopy class of loops $\omega_m \cdot \omega_n$ and we have $\Phi(n + m) = \Phi(n) \cdot \Phi(m)$. So in $\mathbb{R}$, you add $m + n$, means you wind $m + n$ on S^1 . **FINISH PROOF** $\square$

8 Miscellaneous Exercises

Result 8.1. Prove that if $x_1, x_2, \dots, x_k \in \mathbb{R}^m$, then

$$\left\| \sum_{n=1}^k x_n \right\| \leq \sum_{n=1}^k \|x_n\|.$$

Proof. Note that by the triangle inequality

$$\|x_1 + x_2 + x_3\| \leq \|x_1 + x_2\| + \|x_3\| \leq \|x_2\| + \|x_2\| + \|x_3\|.$$

Repeating this process proves the result. □

Result 8.2. Let a and b be real numbers. Then $a \leq b$ if and only if $a < b + \varepsilon$ for every $\varepsilon > 0$.

Proof. In the $\implies$ direction, we assume $a \leq b$. Since $b < b + \varepsilon$ for every $\varepsilon > 0$, we have that $a < b + \varepsilon$.

In the $\impliedby$ direction, we prove by contradiction. Assume that $a < b + \varepsilon$ for every $\varepsilon > 0$ and $a > b$. Let $\varepsilon = \frac{a-b}{2}$. Then

$$a < b + \frac{a-b}{2} = \frac{a+b}{2} < b.$$

Therefore, $a < b$, a contradiction. □

Result 8.3. Prove that for $x \in [0, \frac{\pi}{4}]$,

$$\lim_{n \rightarrow \infty} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} = \cos(x)$$

Proof. We have

$$\cos^n(x) + \sin^n(x) = \cos^n(x) \left[1 + \left(\frac{\sin(x)}{\cos(x)} \right)^n \right]$$

and

$$[\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} = \cos(x) \left[1 + \left(\frac{\sin(x)}{\cos(x)} \right)^n \right]^{\frac{1}{n}}.$$

If $x \in [0, \frac{\pi}{4}]$, then $0 \leq \frac{\sin(x)}{\cos(x)} \leq 1$ and by the Squeeze theorem,

$$1 \leq \left[1 + \left(\frac{\sin(x)}{\cos(x)} \right)^n \right]^{\frac{1}{n}} \leq 2^{\frac{1}{n}}$$

implies

$$\lim_{n \rightarrow \infty} \left[1 + \left(\frac{\sin(x)}{\cos(x)} \right)^n \right]^{\frac{1}{n}} = 1.$$

Therefore,

$$\lim_{n \rightarrow \infty} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \cos(x) \left[1 + \left(\frac{\sin(x)}{\cos(x)} \right)^n \right]^{\frac{1}{n}} = \cos(x).$$

□

Result 8.4. *Prove*

$$\lim_{n \rightarrow \infty} \int_0^{\frac{\pi}{4}} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} dx = \frac{\sqrt{2}}{2}.$$

Proof. Note that

$$\int_0^{\frac{\pi}{2}} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} dx = \int_0^{\frac{\pi}{4}} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} dx.$$

We factor out a cosine term to obtain

$$\int_0^{\frac{\pi}{4}} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} dx = \int_0^{\frac{\pi}{4}} \cos(x) \left[1 + \frac{\sin^n(x)}{\cos^n(x)} \right]^{\frac{1}{n}} dx.$$

Using the dominated convergence theorem and the previous result, we have that

$$\lim_{n \rightarrow \infty} \int_0^{\frac{\pi}{4}} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} dx = \int_0^{\frac{\pi}{4}} \lim_{n \rightarrow \infty} [\cos^n(x) + \sin^n(x)]^{\frac{1}{n}} dx = \int_0^{\frac{\pi}{4}} \cos(x) dx = \frac{\sqrt{2}}{2}.$$

□

Result 8.5. *Let $r \in \mathbb{R}$ with $r \geq 2$ and let $b_1, \dots, b_n \in \mathbb{R}^+$ satisfying $\sum_{i=1}^n b_i = n$. Prove*

$$\sum_{i=1}^n \frac{1}{b_i^r - b_i + n} \leq 1$$

with equality if and only if $b_i = 1$ for all i .

Proof. Finish Proof

□

Result 8.6. *Prove*

$$\lim_{j \rightarrow \infty} \left(\lim_{k \rightarrow \infty} (\cos(j! \pi x))^{2k} \right) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

for every $x \in \mathbb{R}$.

Proof. Fix j and let $x \in \mathbb{R}$. Since

$$0 \leq (\cos(j! \pi x))^2 \leq 1$$

we have that

$$\lim_{k \rightarrow \infty} (\cos(j! \pi x))^{2k} = \begin{cases} 1 & \text{if } j!x \in \mathbb{Z} \\ 0 & \text{if } j!x \notin \mathbb{Z}. \end{cases}$$

Let $x \in \mathbb{Q}$. For some $p, q \in \mathbb{Z}$, we have $x = \frac{p}{q}$ and for some $j \geq q$, q divides $j!$. Therefore, as $j \rightarrow \infty$, we have that $j!x \in \mathbb{Z}$ and

$$\lim_{j \rightarrow \infty} \left(\lim_{k \rightarrow \infty} (\cos(j! \pi x))^{2k} \right) = 1.$$

Let $x \in \mathbb{R} \setminus \mathbb{Q}$. As $j \rightarrow \infty$ we have that $j!x \notin \mathbb{Z}$, thus

$$\lim_{j \rightarrow \infty} \left(\lim_{k \rightarrow \infty} (\cos(j! \pi x))^{2k} \right) = 0$$

proving the result.

□

Result 8.7. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function. Prove that the inverse function $f^{-1} : f(\mathbb{R}) \rightarrow \mathbb{R}$ is a continuous function.

Proof. We fix $x_0 = f^{-1}(y_0)$ and let $\varepsilon > 0$. Let

$$x_0 - \varepsilon < x_0 < x_0 + \varepsilon.$$

Since f is strictly increasing, then

$$f(x_0 - \varepsilon) < y_0 < f(x_0 + \varepsilon).$$

Choose $\delta = \min\{y_0 - f(x_0 - \varepsilon), f(x_0 + \varepsilon) - y_0\}$. Then for any $y \in f(\mathbb{R})$ with $|y - y_0| < \delta$

$$f(x_0 - \varepsilon) < y < f(x_0 + \varepsilon).$$

We have that

$$x_0 - \varepsilon < f^{-1}(y) < x_0 + \varepsilon$$

proving the result. □

To prove the next result, we state the dominated convergence theorem.

Theorem 8.8. Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $f : X \rightarrow [-\infty, \infty]$ is an $\mathcal{S}$ -measurable, and $f_1, f_2, \dots$ are $\mathcal{S}$ -measurable functions from X to $[-\infty, \infty]$ such that

$$\lim_{k \rightarrow \infty} f_k(x) = f(x)$$

for almost every $x \in X$. If there exists an $\mathcal{S}$ -measurable function $g : X \rightarrow [0, \infty]$ such that

$$\int g \, d\mu < \infty \quad \text{and} \quad |f_k(x)| \leq g(x)$$

for every $k \in \mathbb{Z}^+$ and almost every $x \in X$, then

$$\lim_{k \rightarrow \infty} \int f_k \, d\mu = \int f \, d\mu.$$

Result 8.9 (MIRA 3B.5). Let λ denote Lebesgue measure on $\mathbb{R}$. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel measurable function such that $\int |f| \, d\lambda < \infty$. Prove that

$$\lim_{k \rightarrow \infty} \int_{[-k, k]} f \, d\lambda = \int f \, d\lambda.$$

Proof. For each $k \in \mathbb{Z}^+$, define

$$f_k(x) := \chi_{[-k, k]}(x) f(x).$$

For every $x \in \mathbb{R}$, there exists $K \in \mathbb{Z}^+$ such that $x \in [-k, k]$ whenever $k \geq K$. Hence,

$$\lim_{k \rightarrow \infty} f_k(x) = \lim_{k \rightarrow \infty} \chi_{[-k, k]}(x) f(x) = f(x).$$

Moreover,

$$|f_k(x)| = \chi_{[-k,k]}(x)|f(x)| \leq |f(x)|$$

for every $k \in \mathbb{Z}^+$ and every $x \in \mathbb{R}$. Since

$$\int_{\mathbb{R}} |f| d\lambda < \infty,$$

the dominated convergence theorem gives

$$\begin{aligned} \lim_{k \rightarrow \infty} \int_{[-k,k]} f d\lambda &= \lim_{k \rightarrow \infty} \int_{\mathbb{R}} \chi_{[-k,k]} f d\lambda \\ &= \lim_{k \rightarrow \infty} \int_{\mathbb{R}} f_k d\lambda \\ &= \int_{\mathbb{R}} f d\lambda. \end{aligned}$$

□